UNIVERSITY OF NOTRE DAME | COLLEGE OF ENGINEERING

ROAR ROBOTICS AND AUTONOMY RESEARCH LAB

Slide Title Goes Here

Your name

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Body slide example

- Body text renders in Century-style serif (`fouriernc`) to match IEEE LaTeX paper bodies.
- Frame titles render in condensed sans (Barlow Condensed if installed, Helvetica fallback) – substitute for Acumin Pro ExtraCondensed.
- Frame number sits bottom-right, bare integer, no “/N”.

IEEE-style equations

Equations are part of the sentence flow and end with the appropriate punctuation. Define the posterior over the stochastic latent as

$$s_t \sim q_\phi(s_t \mid h_t, o_t), \tag{1}$$

and decompose the world-model objective into a reconstruction term and a KL term,

$$\mathcal{L}_{\text{rec}} = -\mathop{\mathbb{E}}_{q_\phi}\left[\log p_\theta(o_t \mid h_t, s_t)\right], \tag{2}$$

$$\mathcal{L}_{\text{KL}} = \mathrm{KL}\big(q_\phi(s_t \mid h_t, o_t) \,\big\|\, p_\theta(s_t \mid h_t)\big). \tag{3}$$

Combining (2) and (3) with (1) gives the ELBO used during training.

Inline conventions: bold lowercase italic for vectors $\boldsymbol{x} \in \mathbb{R}^n$, bold capital italic for matrices $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, upright roman subscripts for word labels ($\mathcal{L}_{\text{rec}}$ not $\mathcal{L}_{rec}$), optimizers via $\arg\min_\theta \mathcal{L}(\theta)$.